

$$\frac{\beta_p}{\beta_n} = \frac{k_p}{k_n}$$

$$V_{IT} = \frac{V_{DD} - |V_{TP}| + \sqrt{\frac{k_n}{k_p}} V_{TN}}{1 + \sqrt{\frac{k_n}{k_p}}}$$

$$\frac{k_n}{k_p} \uparrow, V_{IT} \downarrow$$

$$\frac{k_n}{k_p} \downarrow, V_{IT} \uparrow$$

① $\frac{\beta_p}{\beta_n} > 1$ High skewed CMOS inverter.
Strong PMOS.

$$\frac{R_n}{R_p} > 1$$

$$\begin{aligned} \beta_n &\propto \frac{1}{R_n} \\ \beta_p &\propto \frac{1}{R_p} \end{aligned}$$

$R_n > R_p$ } Not symmetrical.
↓
nmos Resistance more.

slow discharge →
fast charge.

T_{PLH} fast,

$$T_{PLH} > T_{PHL}$$

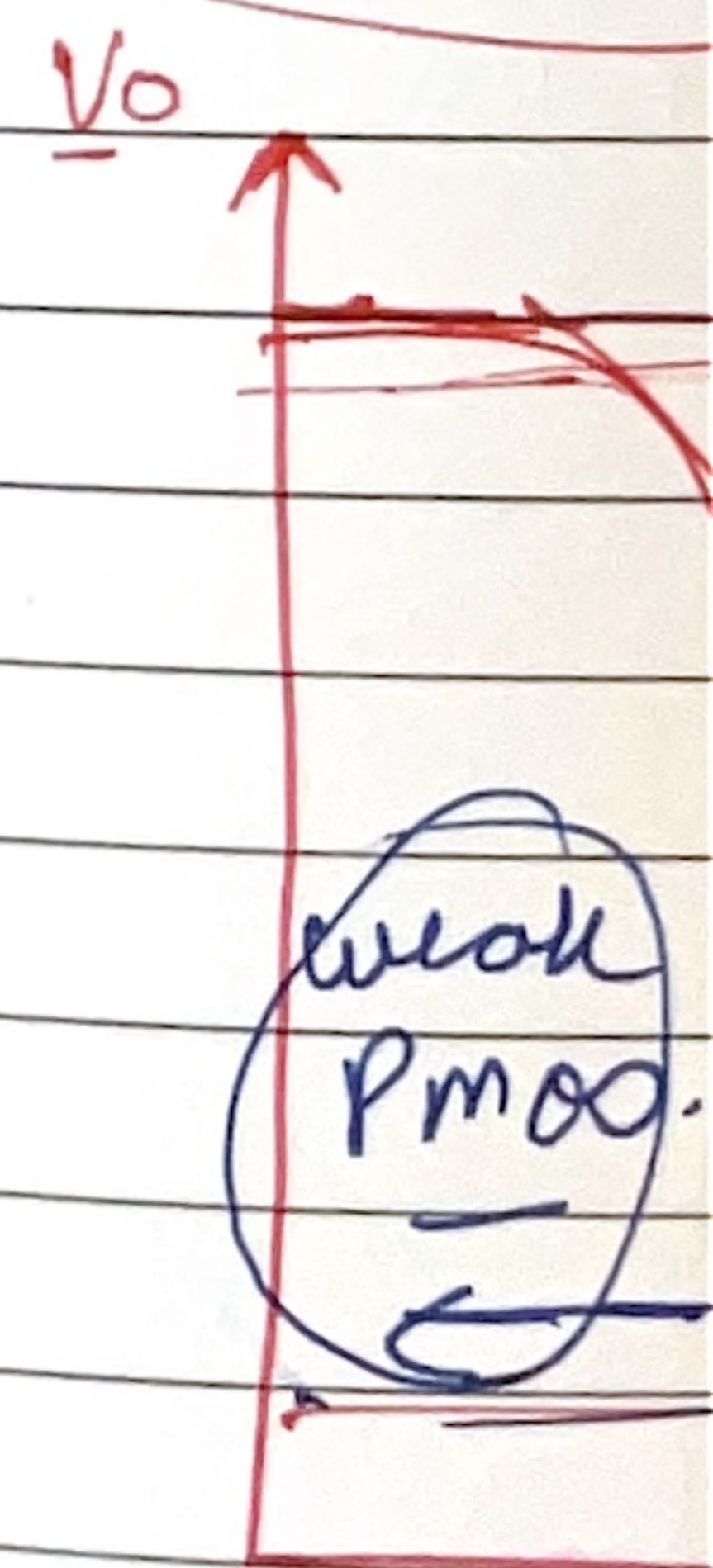
$$\textcircled{2} \frac{\beta_p}{\beta_n} < 1$$

$R_n < R_p$
↑
fast discharge

$$T_{PLH}$$

$$\textcircled{3} \frac{\beta_p}{\beta_n} = 1$$

$$T_{PLH} = T_{PHL}$$



$$\textcircled{1} \frac{\beta_p}{\beta_n} > 1$$

$$\textcircled{2} \frac{\beta_p}{\beta_n}$$

Case II

- ② $\frac{\beta_p}{\beta_n} < 1$ — low skewed CMOS inverter
— Weak Pmos.

$R_n < R_p$
↑ fast discharge
↑ slow charge.

$$T_{PLH} < T_{PHL}$$

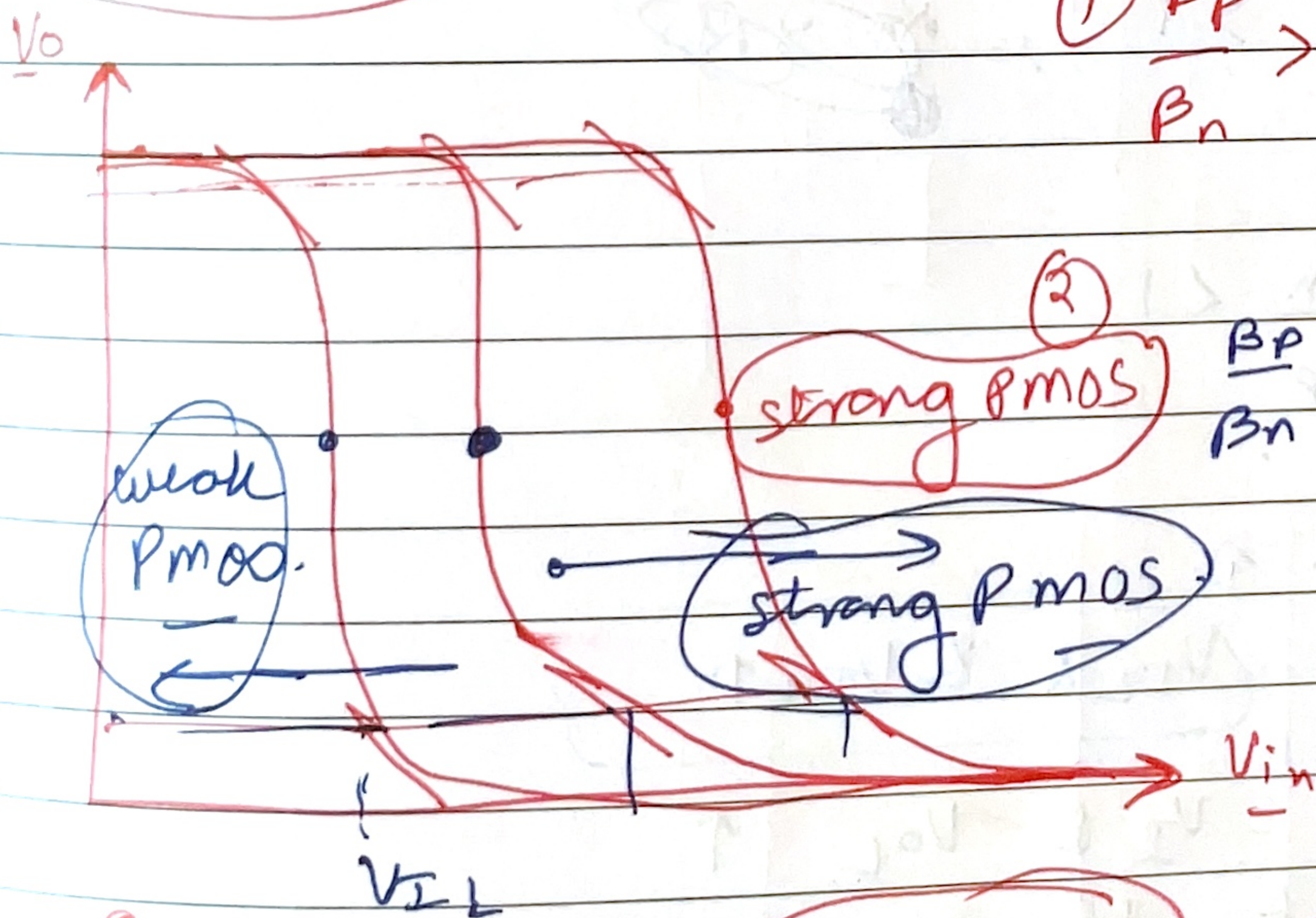
We want it to be equal kind of.

- ③ $\frac{\beta_p}{\beta_n} = 1$ | $\frac{R_n}{R_p} = 1$ [V_T, V_{GS} & Remaining all are equal.]

$$T_{PLH} = T_{PHL}$$

$$\frac{\mu_n C_{ox} \left(\frac{W}{L} \right)}{\mu_p C_{ox} \left(\frac{W}{L} \right)} \left(\frac{V_{GS} - V_{TP}}{V_{TN}} \right)^2 = 1$$

① $\frac{\beta_p}{\beta_n} > 1$ $R_n > R_p$ (strong PMOS)



- ④ $\frac{k_p}{k_n} > 1$, $\frac{k_n}{k_p} < 1$ — strong PMOS.

- ⑤ $\frac{\beta_p}{\beta_n} < 1$ or $\frac{k_p}{k_n} < 1$ or $\frac{k_n}{k_p} > 1$ or, $R_n < R_p$
 $V_{IT} \uparrow$
 $V_{IT} \downarrow$ fast discharge

When $\frac{k_p}{k_n}$ changes, V_{IH} & V_{IL} increases/decreases
 and V_{OH} & V_{OL} are un-affected -

→ Equal Rise & Fall time, symmetrical CMOS
 $B_n = B_p$
 $B_n = B_p$
 $R_n = R_p$

- width of the PMOS transistor is increased
 $(\frac{W}{L})_p \uparrow$ effect on Low Noise Margin
 High Noise Margin

Now $B_n = B_p$ is no more valid.

~~$\frac{B_p}{B_n} > 1$~~ , ~~$R_n > R_p$~~

$\frac{k_n}{k_p} < 1$

~~R_n~~ $V_{IT} \uparrow$

$V_{IL} \downarrow$ & $V_{IH} \uparrow$

$N_{MH} = V_{IL} - V_{OL} \uparrow$

$N_{ML} = V_{OH} - V_{IH} \downarrow$

only V_{IN} & V_{OL} changes.

→ $NMH = NML$

$R_n = R_p$ - equal propagation delay.

$$R_n = \frac{1}{\mu_n \left(\frac{w}{L} \right) (V_{GS} - V_{TN})} = \frac{1}{k_n (V_{GS} - V_{TN})}$$

$$R_p = \frac{1}{\mu_p \left(\frac{w}{L} \right) (V_{GS} - V_{TP})} = \frac{1}{k_p (V_{GS} - V_{TP})}$$

for Symmetric $R_n = R_p$.

$$k_n = k_p, \quad V_{TN} = |V_{TP}|$$

Practically $\mu_n \neq \mu_p$ →
 $\mu_n \approx 2\mu_p$

$$\mu_n \left(\frac{w}{L} \right)_n \quad \mu_p \left(\frac{w}{L} \right)_p$$

$L \propto$ sub threshold conduction

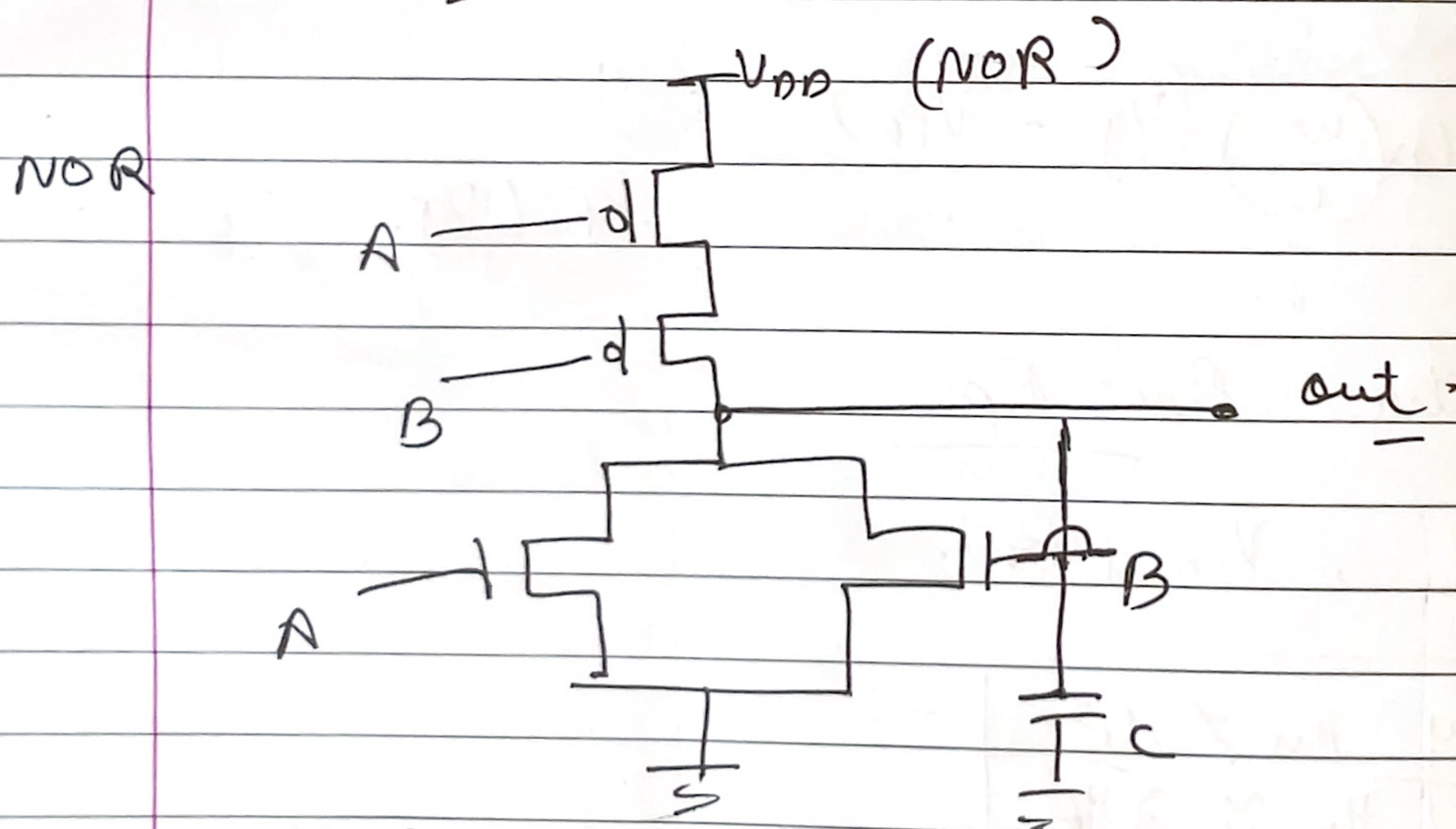
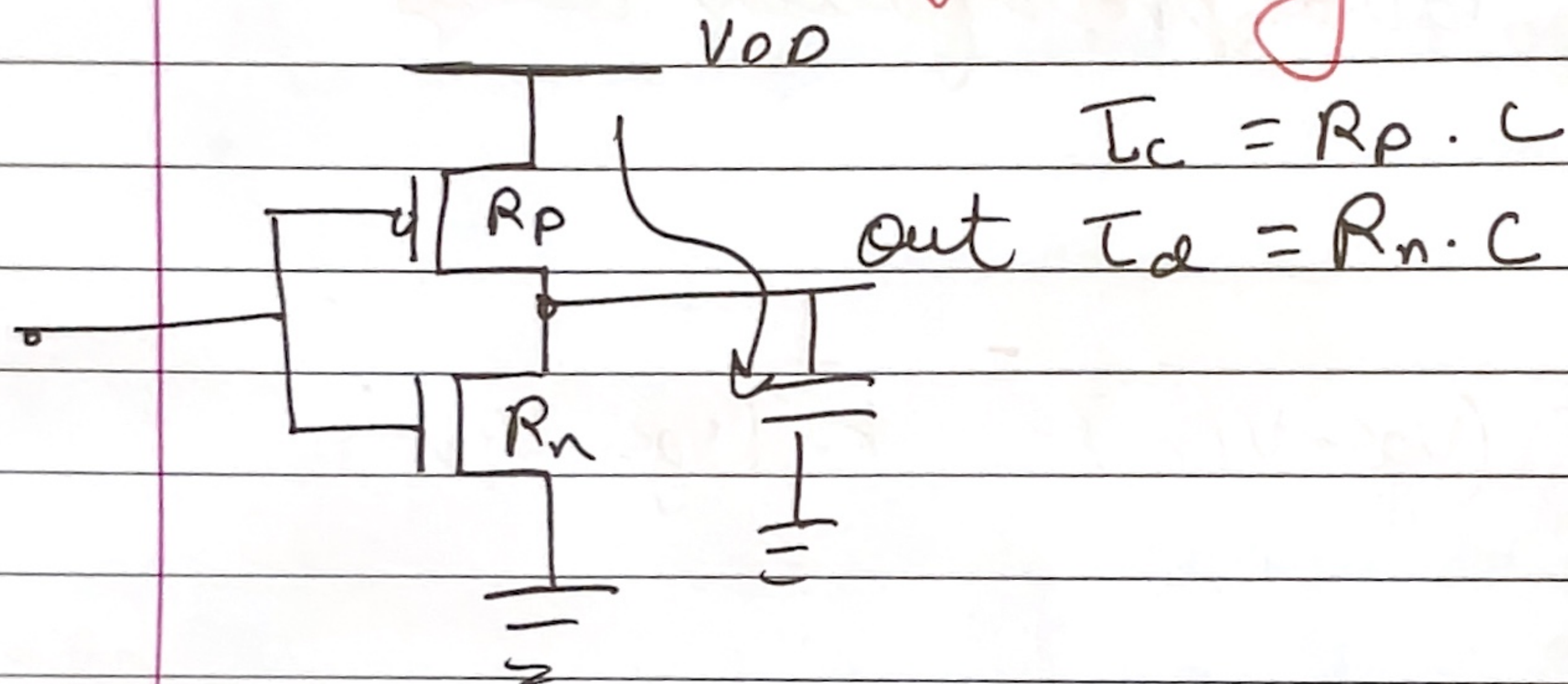
$$\left(\frac{w}{L} \right)_p = 2 \left(\frac{w}{L} \right)_n \quad \text{then we will get } k_n = k_p,$$

$$R_n = R_p$$

for symmetry - I need large PMOS.

$$\left(\frac{W}{L}\right)_p = 2 \left(\frac{W}{L}\right)_n$$

To maintain symmetry.



NOR

A	B	out
0	0	1
0	1	0
1	0	0
1	1	0

① PMOS: ON
 NMOS: OFF

$$T_c = 2 R_p' \cdot C$$

② 0 1

$$T_d = R_n' \cdot C$$

③

$$T_d = R_n' \cdot C$$

④

$$t_d = \frac{R_n' \cdot C}{2}$$

charging

- Worst / Best Case

- ~~2 R_p~~ 2 R_p' C

- Worst Case discharging

= R_n' C

$R_p \cdot C = 2 R_p' \cdot C$

$R_p = 2 R_p'$

$R_p' = \frac{R_p}{2}$

$R_p \propto \frac{1}{\omega_p}$

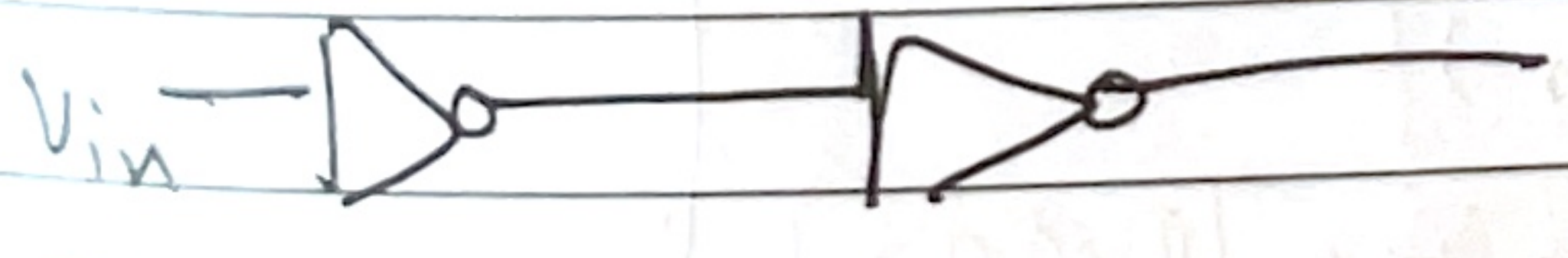
$\frac{1}{\omega_p} = \frac{1}{2\omega_p'}$

$\omega_p' = 2\omega_p$

→ for discharging

$R_n \cdot C = R_n' \cdot C$

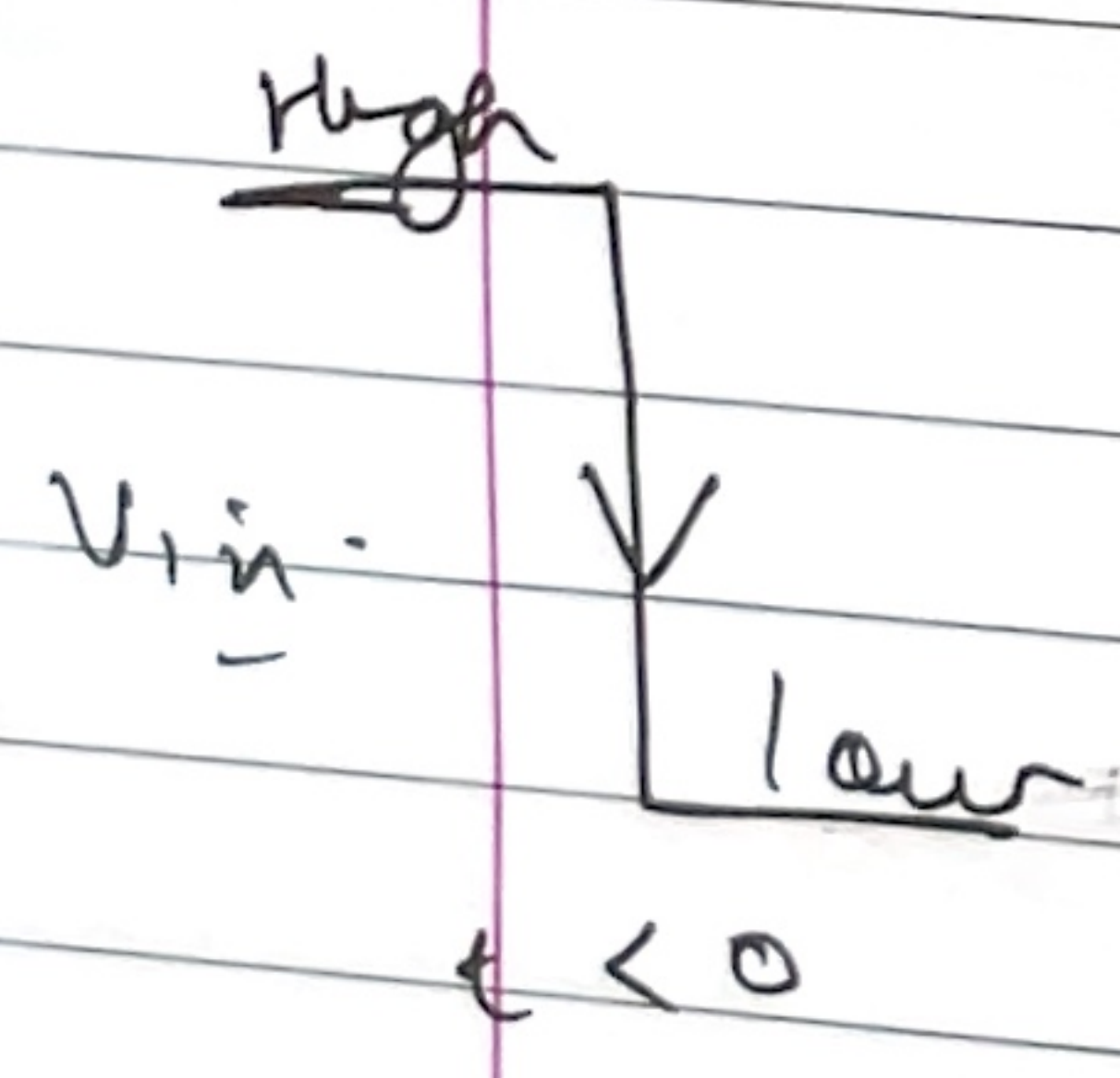
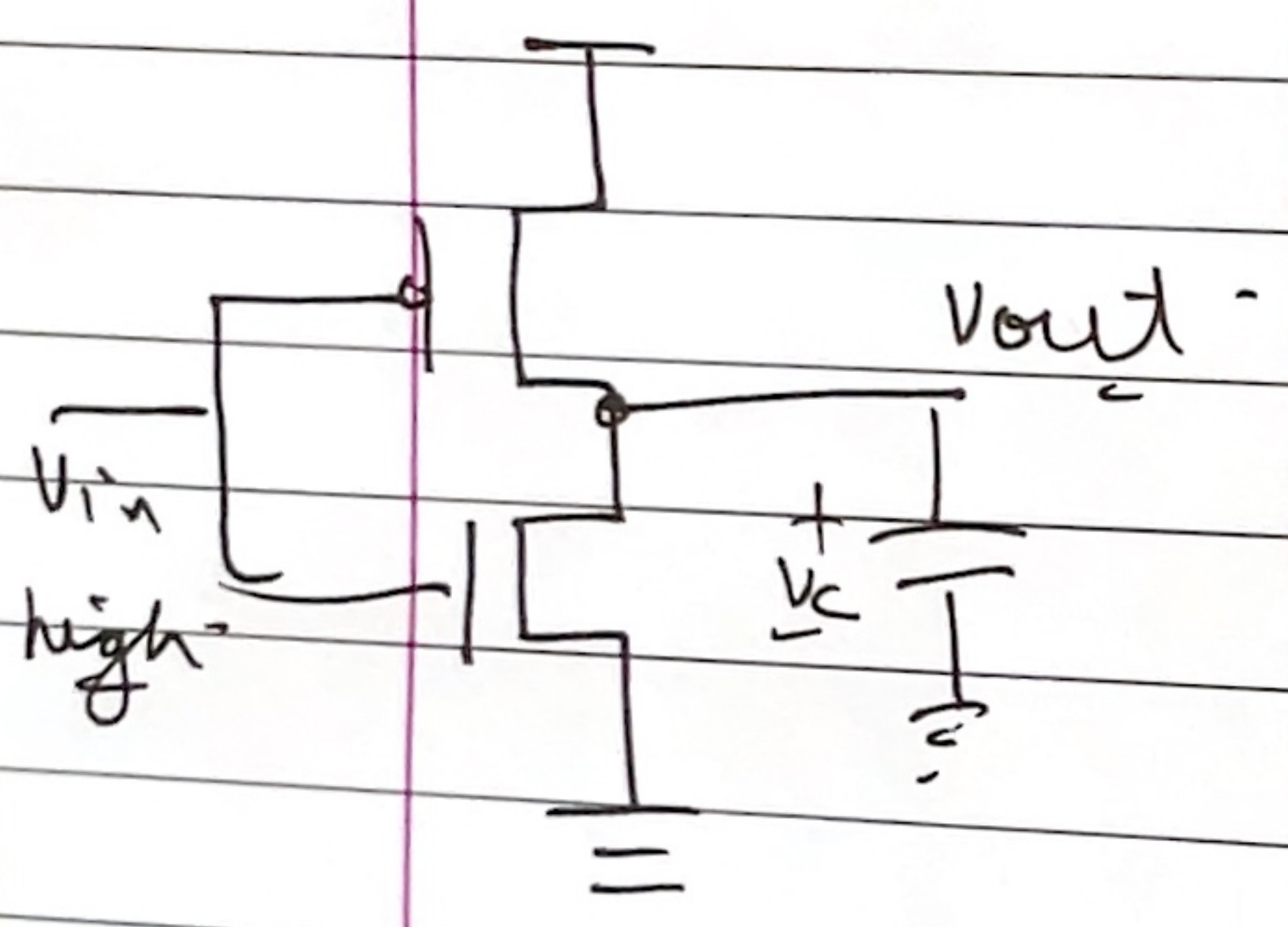
Propagation delay.



T_{PLH}
 $t < 0$

$t = 0^+$

$t \geq 0$



NMOS low

N : cutoff

P : linear

$$Q_c(\infty) = C \cdot V_{DD}$$

$$V_c(\infty) = V_{DD}$$

NMOS $\rightarrow$ linear

PMOS $\rightarrow$ cutoff

1) P high

$$V_c(0^-) = 0V$$

$$Q_c(0^-) = 0V$$

$0^+ \rightarrow$ high to low transition

PMOS $\rightarrow$ cutoff $\rightarrow$ linear
 Saturation

NMOS - linear

$$V_c(0^+) = 0$$

$$Q_c(0^+) = 0$$

Saturation (PMOS)

$\rightarrow$ linear
 PMOS

Average Charge

$$T_{PLH} = \frac{Q_c}{I_D}$$

$$\therefore Q = it$$

$$t_{PLH} = Q_c(\infty) = CV_{DD}$$

Average 150% Rule.

$$Q_c = 0 + \frac{CV_{DD}}{2} = \frac{1}{2} CV_{DD}$$

$$T_{PLH} = \frac{1}{2} CV_{DD} \therefore$$

$$\text{sat} \quad \frac{1}{2} \mu_p \left(\frac{W}{L} \right)_p (V_{GS} - V_{TP})^2$$

$$t_{PLH} = \frac{CV_{DD}}{\mu_n \left(\frac{W}{L} \right)_p (V_{GS} - V_{TP})^2}$$

V_{DD} , T_{PLH}

$V_{DD} \uparrow$, $T_{PLH} \downarrow$ (fast charge) ✓

→ $V_{DD} \uparrow$, $T_{PLH} \downarrow$

→ $(V_{GS} - V_{TP})^2$
 $(V_{in} - V_{DD} - V_{TP})^2$

$$V_{TP} = -1V$$

$$V_{in} = 2V$$

$$P_y \quad V_{TP} = -1V$$

$$V_{in} = 2V$$

$$V_{DD} = 4V$$

$$V_{DD} = 5V$$

$$V_{DD} = 4V$$

$$V_{DD} = 5V$$

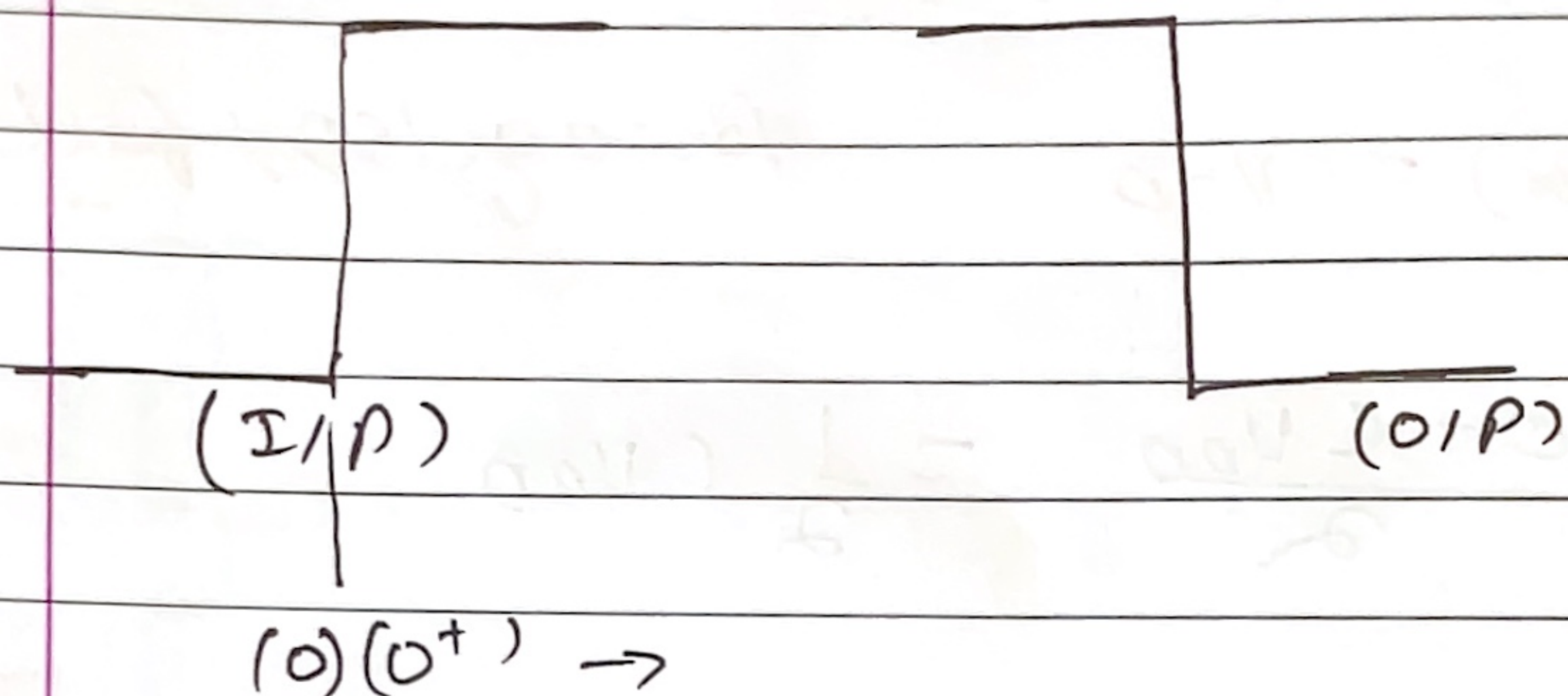
pmos on/saturation

$V_{DD} \uparrow$, $I_n \uparrow$, fast charge.

$$V_{DD} = 4$$

$V_{DD} \uparrow$, $T_{PLH} \downarrow$

T_{PHL} o/p going from high to low.



at $t = (0^-)$	$t = (0^+)$	$t \geq 0$
steady state	transition	
Pmos : on	linear	cut off
Nmos : off (cut off)	saturation	linear
Capacitor charged		discharge through NMOS
$V_o(t < 0) = V_{DD}$		
$Q_c(0^-) = C \cdot V_{DD}$	$Q_c(0^+) = C \cdot V_{DD}$	
	→ NMOS in (sat)	→ NMOS in sat
		→ most of the NMOS will discharge through NMOS in saturation condition

$T_{PHL} = \frac{Q_c}{(I_{DN})_{NMOS \text{ current in saturation}}}$ (Average 50% cap. charge)

$$= \frac{\frac{1}{2} C V_{DD}}{\frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right) (V_{GS} - V_{TN})^2} = \frac{C V_{DD}}{\mu_n C_{ox} \left(\frac{W}{L}\right) (V_{GS} - V_{TN})^2}$$

V_{DD} will be $\uparrow$

T_{PHL} will $\uparrow$

Cause $V_{GS} - V_{TH}$

$= V_{in} - V_{TH}$ no dependence on V_{DD} .

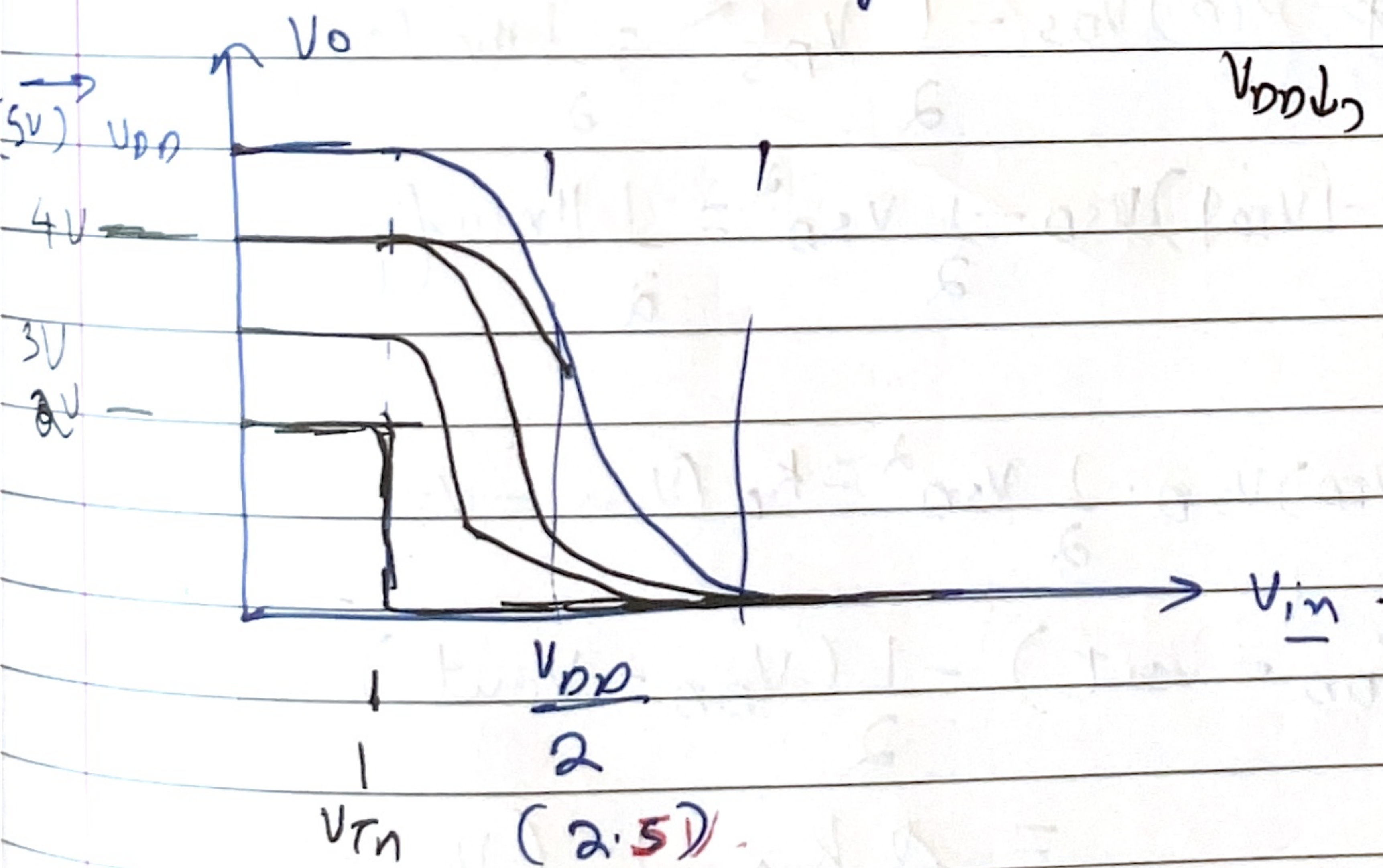
$T_p = \overset{\text{will } \uparrow}{t_{PHL}} + \overset{\text{will } \downarrow}{t_{PLH}}$

so what will be the impact on Propagation delay will decrease.

① $V_{DD} \uparrow$, $T_p \downarrow$

② $P_{dynamic} = C_{in} \cdot C \cdot V_{DD}^2$

③ $V_{DD} \uparrow$, $T_p \downarrow$, $P_{dynamic} \uparrow$



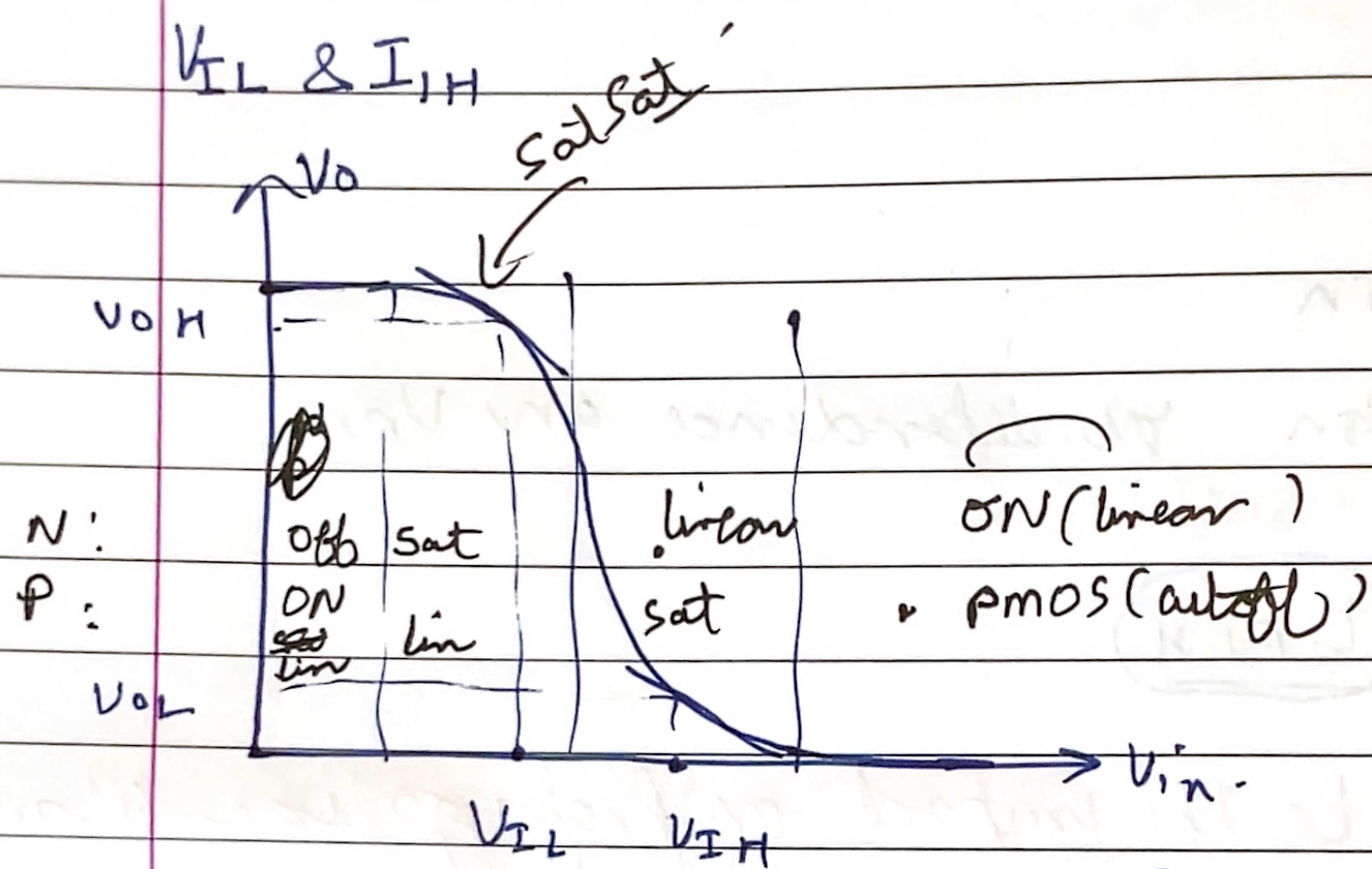
$V_{DD} \downarrow$, $T_p \uparrow$ (high propagation delay).

(Ideal inverter)

① $0 < V_{in} < 1V \rightarrow$ NMOS cutoff, PMOS linear ($V_O = 2V$)

② $1 < V_{in} < 2V \rightarrow$ NMOS $\rightarrow$ linear, PMOS $\rightarrow$ cutoff

No saturation state.



$$V_O = B(V_{in})$$

V_{IL} : NMOS: sat
PMOS: linear

$$I_{OP} = I_{ON}$$

→ NMOS sat

$$\mu_p C_{ox} \left(\frac{W}{L} \right)_p (V_{gs} - V_{TP}) V_{DS} - \frac{1}{2} V_{DS}^2 = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_n (V_{gs} - V_{TN})^2$$

$$\mu_p C_{ox} \left(\frac{W}{L} \right)_p (V_{sg} - |V_{TP}|) V_{SD} - \frac{1}{2} V_{SD}^2 = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_n (V_{gs} - V_{TN})^2$$

$$K_p (V_{DD} - V_{in} - |V_{TP}|) V_{SD} - \frac{1}{2} V_{SD}^2 = K_n (V_{in} - V_{TN})^2$$

$$K_p (V_{DD} - V_{in} - |V_{TP}|) (V_{DD} - V_{out}) - \frac{1}{2} (V_{DD} - V_{out})^2 = \frac{1}{2} K_n (V_{in} - V_{TN})^2$$

→ different V_{in}

$$\Rightarrow V_{IL} = \frac{1}{8} [3V_{DD} + 2V_{TN}]$$

V_{TN}
 V_{IH}

N: linear
P: sat

$$k_n = K_n$$

$$\frac{1}{2} K_p (V_{gs} - V_{TP})^2$$

$$V_{IH} = \frac{1}{8} [3V_{DD} + 2V_{TN}]$$

$V_{IL} + V_{IH}$

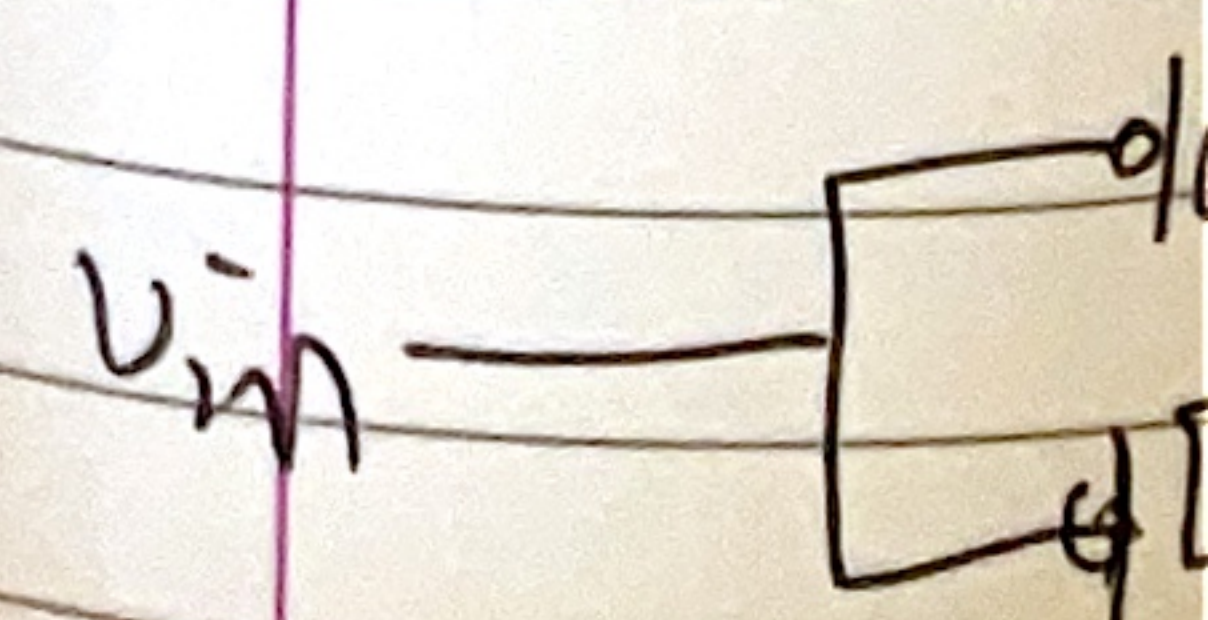
$$N_{ML} =$$

$$V_{IL} =$$

$$V_{MN} = V_{DD} - V_{IL}$$

$$V_{MN} = V_{DD} - V_{IL}$$

→ DO



V_{Tn}

V_{IH}

N: Linear

P: Saturation

$$\frac{dv_o}{dv_{in}} = -1$$

$k_n = k_p$ pMOS $\longleftrightarrow$ nMOS

$$\frac{1}{2} k_p (V_{GS} - V_{TP})^2 = k_n (V_{GS} - V_{Tn}) V_{DS} - \frac{1}{2} V_{DS}^2$$

$$V_{IH} = \frac{1}{8} [5V_{DD} - 2V_{Tn}]$$

$$V_{IL} + V_{IL} =$$

$$N_{ML} = V_{IL} - V_{OL} = V_{IL} - 0$$

$$V_{IL} = V_{DD} - V_{IH}$$

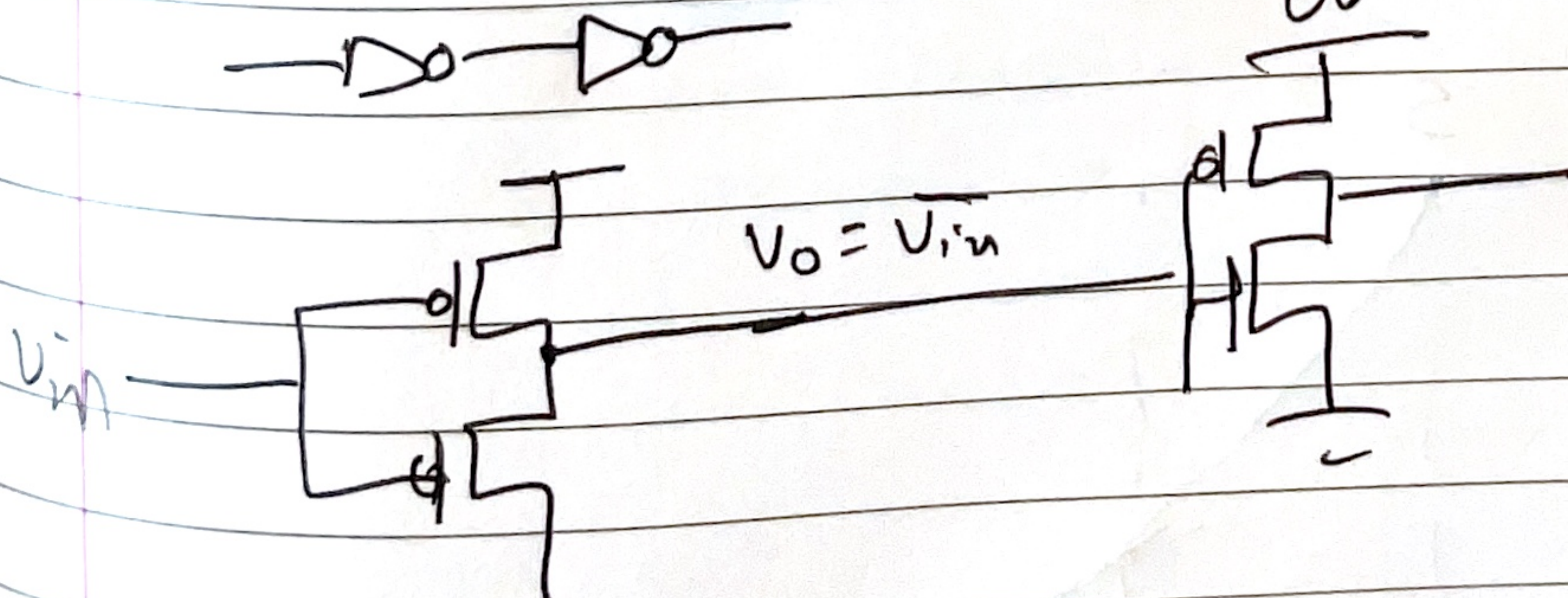
$$V_{Mn} = V_{OH} - V_{IH}$$

$$V_{Mn} = V_{DD} - V_{IH}$$

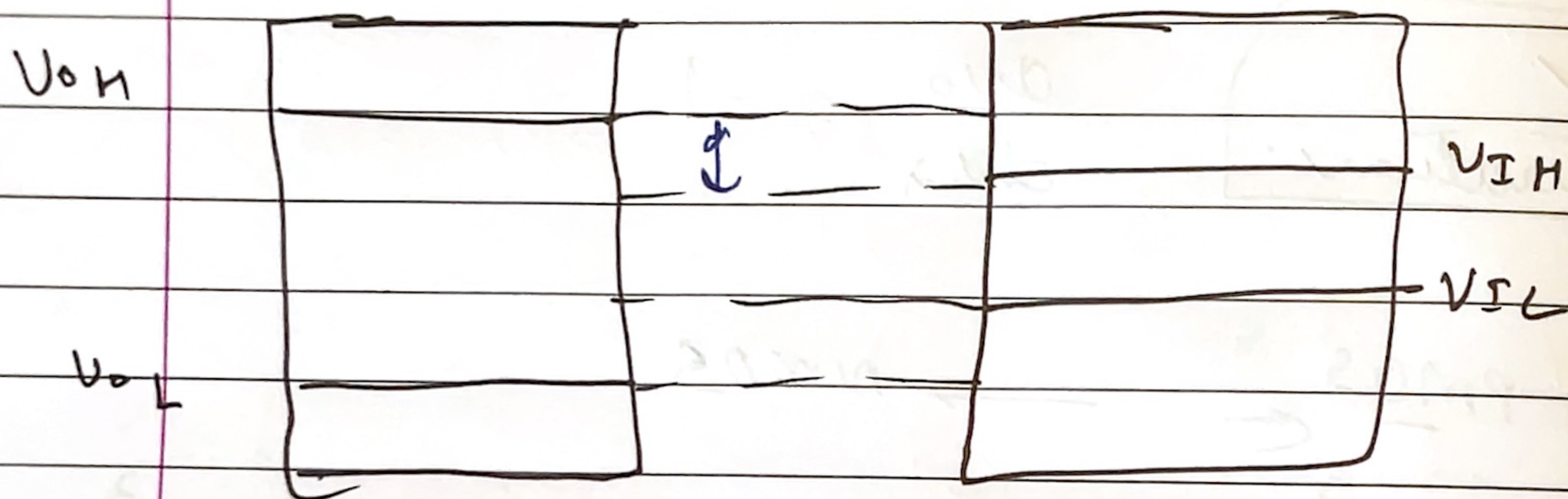
$$= V_{DD} - V_{DD} + V_{IL}$$

$$V_{Mn} = V_{IL}$$

CMOS Buffer



Select the correct statements



$$N_{MH} = V_{oh} - V_{IH}$$

$$N_{ML} = V_{IL} - V_{ol}$$

(c)

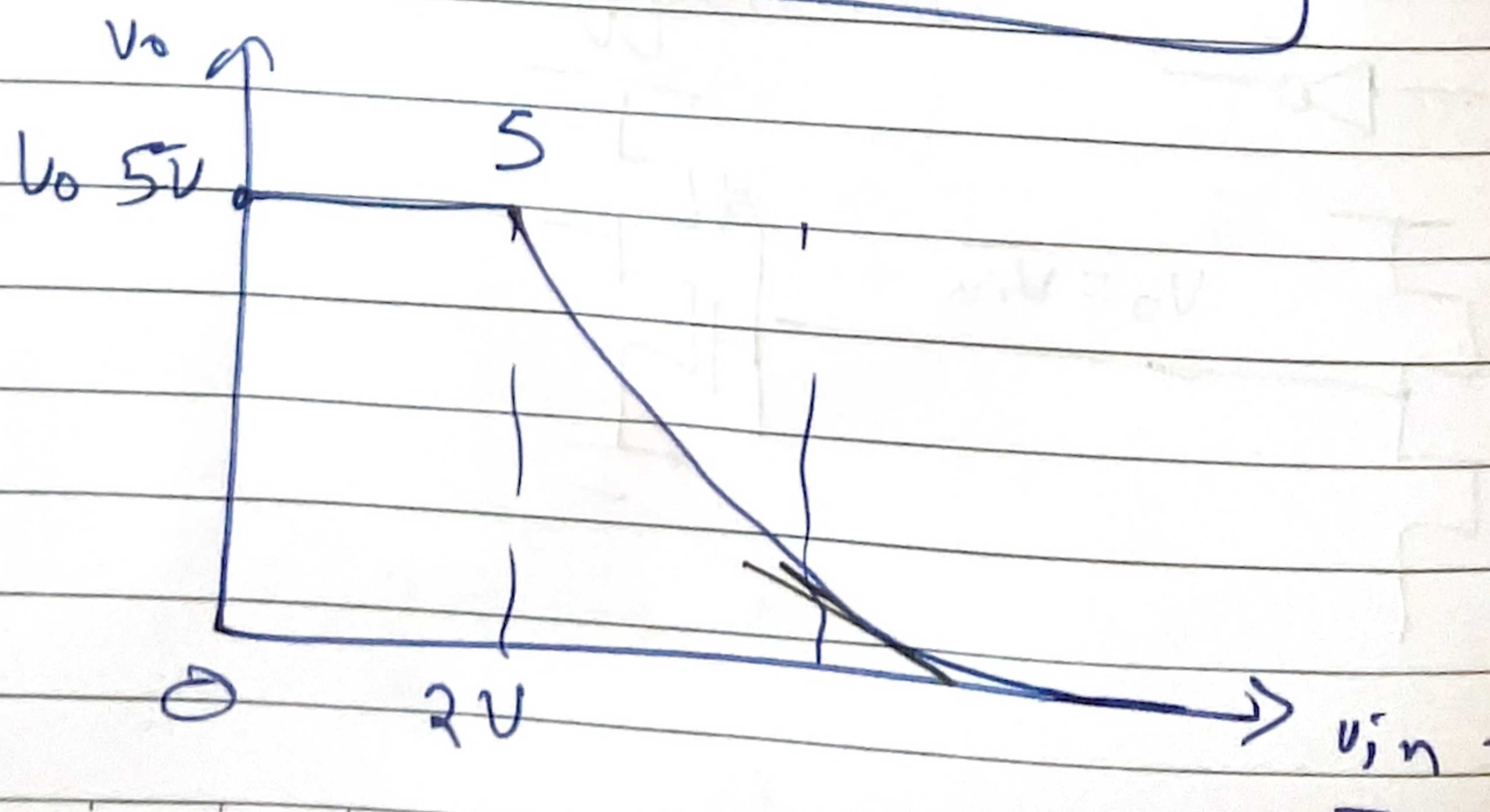
$$T_{PLH} = \frac{1}{2} C V_{DD}$$

$$\frac{1}{a} \mu_p \left(\frac{W}{L} \right)_p (V_{GS} - V_{TP}) (I_{Dsat})$$

sat current of PMOS.

$$= \frac{C V_{DD}}{\mu_p \left(\frac{W}{L} \right)_p (V_{GS} - V_{TP})^2}$$

Q.



$$5\mu (3 - 1 \cdot 5V_{in})$$

$$V_{on} \approx 5V$$

$$V_{o2} \approx 0V$$

$$V_{in} \approx$$

$$(3 - 1 \cdot 5V_{in}) = -1$$

$$\frac{d}{dV_{in}} 5\mu$$

$$3 - 1 \cdot 5V_{in}$$

$$5\mu \cdot (-1 \cdot 5) = -1$$

$$7.5\mu \cdot (3 - 1 \cdot 5V_{in}) = 1$$

$$7.5\mu \cdot (3 - 1 \cdot 5V_{in}) = 1 \quad \left| \quad V_{in} = \underline{V_{IH}} \right.$$

$$\frac{3 - 1 \cdot 5V_{in}}{7.5} = 1$$

$$V_{IH} \rightarrow \log$$